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Developing an ADL profile for bipolar disorder: A systematic review protocol

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Developing an ADL profile for bipolar disorder: A systematic review protocol

ABSTRACT

Introduction: Bipolar disorder (BD) is a severe mental health disorder that affects 2% of the adult population. Individuals with BD are at a higher risk for morbidity and functional impairment. They may also experience significant challenges in their activities of daily living (ADLs). There is a need to better understand the implications of BD in managing ADLs. This systematic review is planned to identify all available studies that examine ADLs that are impacted in individuals with BD

Methods and analysis: Medline, Embase, CINAHL, and APA PsycINFO will be searched to identify observational studies that examined BD patients' functioning, independence, or performance across individual ADLs or subsets of ADLs. Title and abstract, full-text screening, and a risk of bias assessment will be conducted in duplicate. An overarching 'ADL profile' or table summarizing the available studies will be developed. The table will be organized with each specific ADL that is identified and indicate the BD patient's functioning or independence.

Ethics and dissemination: As this systematic review uses information from previous literature, this review does not require ethics approval. This review will help identify the trends in ADLs that individuals with BD struggle with and can help health care practitioners identify specific areas of need for interventions. We plan to disseminate the results in a peer reviewed journal and conferences targeting occupational therapists, and mental health clinicians.

Registration details: This systematic review has been registered on the PROSPERO International Prospective Register of systematic with registration number CRD42021255089.

Strengths and limitations of this study

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- This study will be the first systematic review to examine how performance in ADLs across individuals with BD, and where they typically are most independent.
- This review has an established protocol for assessing risk of bias to ensure that study quality is considered and made transparent to the reader.
- Many articles related to measuring ADLs will be observational studies that could be prone to bias and confounding data, which may limit conclusions to the general BD population.

INTRODUCTION

Bipolar disorder (BD) is a severe and chronic mental health disorder characterized by recurrent mood episodes ranging from depressive to manic states and changes in energy levels.[1] There are two major forms of BD, including bipolar I disorder (BD-I), which involves a manic episode with a hypomanic episode or depressive episode, whereas bipolar II disorder (BD-II) involves at least a depressive episode paired with a hypomanic episode.[1] BD typically presents in early adulthood and is estimated to have a lifetime prevalence of 0.6% for BD-I, and 0.4% for BD-II worldwide.[2] Lifetime prevalence of BD in the United States was estimated at 1.0% and 1.1% for BD I and II, respectively.[2] Individuals with BD also have an increased risk for comorbid mental health disorders and physical illnesses.[3] They may also be at high risk for premature mortality due to suicide and various medical illnesses such as circulatory or respiratory diseases than the general population.[4,5] From a meta-analysis by Hayes et al. (2015),[4] the standardized mortality ratio (SMR) was 2.05 for all-cause mortality, with suicide and unnatural death comprising the highest SMRs, at 14.44 and 7.42, respectively. BD also has high costs on society. In 2009, the annual cost of BD was estimated at $151 billion USD in the

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United States.[6] The majority of these costs are attributed to indirect costs, such as living in institutions and treatment related to functioning rather than direct treatment of symptoms.[6,7]

BD can greatly disrupt functioning in day to day activities across all life domains, including, but not limited to disruptions in basic functioning, work and social roles.[8–10] BD especially impacts a person's activities of daily living (ADLs), which includes any fundamental activities and routine sets of tasks that are relevant to independent functioning in everyday life, whether it involves self-care, homemaking, work, or leisure.[11] It is often distinguished into two types of ADLs: basic ADLs (BADLs), and instrumental or extended ADLs (IADLs or EADLs).[12,13] BADLs typically involve basic everyday activities needed for independence such as mobilization, feeding, dressing, personal hygiene, and toileting.[14] IADLs or EADLs typically encompasses higher level activities, such as managing transportation, handling finances, shopping, meal preparation, home maintenance, as well as managing medications.[13,14]

Original research studies looking at mental health and functioning have found that BD patients often have significantly decreased functioning and performance in ADLs. Numerous researching studies have found significantly decreased functioning in ADLs.[9,15–17] Even after remission from BD, studies report decreased ADL performance and participation in the BD population, where 30-60% of individuals never seeing a return to full functioning.[18] Decker et al. (2017) examined 43 patients in remission, and they found that functioning in daily life activities in these patients remained significantly impaired.[19] For 20-50% of those individuals, the dysfunctions manifested through increased physical effort, fatigue or disorganization.[19] Therefore, BD has a significant impact on daily life functioning, even when symptoms are minimal. Also, the recovery trends imply that absence of symptoms does not guarantee restored

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functioning; thus, individuals who are no longer receiving direct care for BD often continue to contribute to high societal costs.[6,7] Therefore, it is important to consider which areas of functioning in day-to-day activities are most impacted to further reduce societal costs.

In current literature, BD is often studied in terms of social functioning and work roles. Studies have found that social functioning is impacted in various ways, where individuals with BD experienced decreased social interactions, and feelings of social discomfort.[20,21] They also may have impacted relationships with family, relatives, and marital relationships.[20,22] Likewise, BD appears to impact work employment, where there is a 40-60% rate of employment, and if employed, they may be more likely to take days off and underperform in their employment.[23,24] Nevertheless, engaging in work and social situations often requires individuals to manage their basic and instrumental ADLs adequately. If an individual's ability to manage their ADLs is impaired, then they may have difficulties meeting and maintaining their social or work roles.

Despite the general interest in ADL functioning across BD diagnoses, it is unknown which ADLs are most involved in BD, or if there are any trends in ADL impairment. A growing number of studies have examined specific ADLs individuals with BD have difficulties with.[19,25,26] There have also been studies examining how functioning across various ADL domains compare to work and social functioning.[9,16,27] However, there is no current systematic review that summarizes the level of independence or ADL impairment that these patients experience when performing in specific ADLs or in more general ADL-domains. This systematic review will be conducted to help inform health care practitioners about the types and subsets of ADLs that BD patients typically struggle with, and the extent of ADL independence or performance that BD patients typically have. Studying the level of independence for specific

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ADLs can help inform practical and therapeutic approaches in therapeutic settings by giving clinicians targeted areas for intervention planning. There are two questions that will be addressed in this upcoming review:

(1) What does the typical ADL profile look like for individuals with a bipolar disorder?

(2) Which ADL-based occupations do individuals with bipolar disorder have the most difficulty performing?

METHODS

Study registration

This systematic review has been registered on the PROSPERO International Prospective Register of systematic with registration number CRD42021255089. The protocol has been prepared using the Preferred Reporting Items for Systematic reviews and Meta-Analyses for Protocols.[28]

Search strategy

All articles will be collected from the following databases: Ovid MEDLINE, Embase, APA PsycINFO, and CINAHL (Ebsco). A search strategy was developed with the aid of a health systems librarian. The search strategies were also reviewed using the PRESS 2015 Guideline Evidence-Based Checklist.[29] In general, the following search terms and variants of these terms will be used to acquire studies: “activities of daily living”, “functional independence”, and “bipolar disorder”. This search strategy incorporated Boolean searching, truncation, and adjacency. Controlled vocabulary terms were also included, such as Medical Subjects Headings (MeSH), CINAHL Subject Headings, and EMBASE’s Emtree. See Appendix A for the search strategy that was created for each database. Due to the risk of missing studies, searching reference lists of other included studies will identify relevant published studies that did not

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appear from the search strategy. The review management software, Covidence will collect all studies that pass the first and second level of screening.[30]

Inclusion criteria

Only original primary articles published in peer-reviewed journals will be included. Primary articles include any observational studies, including case-control, cross-sectional and cohort studies, but if relevant experimental studies are found, they will also be included. The article must be available in English. Non-English articles that appear from the search will be grouped under "Studies Awaiting Classification" in the PRISMA flow diagram to inform readers of the availability of other potentially relevant reports.[28] No restrictions will be placed on the article's year of publication.

With regards to the patient population, the study must include BD patients ages 18 or older. BD patients include individuals who were diagnosed one of the following based on the Diagnostic and Statistical Manual of Mental Disorders (DSM-V or DSM-IV): Bipolar I Disorder, Bipolar II Disorder, Cyclothymic Disorder, and unspecified bipolar and related disorder.[1] This review will also include studies with patients that have bipolar-specific symptoms, such as euthymia or bipolar-related depressive symptoms, or bipolar-related cognitive deficits.

The inclusion criteria for the study also considers the functional performance or independence in activities of daily living (ADLs), including basic and instrumental/extended ADLs. Basic activities of daily living (BADLs) encompass any self-care activities, such as mobility, feeding, dressing, personal hygiene, or toileting. Instrumental ADLs may encompass transportation, managing finances, shopping, meal preparation, household management, or managing medications. To measure functional performance in ADLs, this review may include a variety of measures such as the UCSD Performance-Based Skills Assessment (UPSA),[31]

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Lawton IADL scale [32], Katz Index of Independence in Activities of Daily Living,[33] Performance Assessment of Self-Care Skills (PASS),[34] Functioning Assessment Short Test (FAST),[35] and any other relevant performance-based measures that have been previously validated in adult or older adult populations. Self-reported measures may also be used such as the Activities of Daily Living Questionnaire (ADL-Q).[19] Distinctions will be made between performance-based data and self-reported data.

Exclusion criteria

Studies will be excluded if they do not examine functional performance in terms of ADLs. They will also be excluded if ADL performance or independence is measured across multiple ADLs, meaning that it does not describe the participant's functioning in specific ADLs such as grooming or managing medications. Studies will also be excluded if they solely focus on work or employment or social functioning while disregarding BADLs or IADLs, as there are currently studies on social functioning, work, and relevance to BD.

Studies will be excluded if they report ADL performance or functioning globally across other mental health disorders, such as reporting ADL performance jointly with schizophrenia or major depressive disorder. Studies examining populations with BD and comorbidities will be excluded. Examples of comorbidities are neurological conditions, cancer diagnoses, substance abuse disorders, developmental delays, or other mental health disorders. Comorbidities are excluded, as studies that investigate comorbidities may also influence ADL performance. Likewise, studies that measure pharmacological or non-pharmacological treatments will be excluded as the purpose of this review is to examine ADLs rather than medical effects.

Screening and selection process

Study selection will occur in duplicate and conducted with two screening levels to

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determine if the articles meet the inclusion criteria. For the first level of screening, two reviewers will independently conduct the search strategy and review all the titles and abstracts from electronic database searches to screen for relevant studies. For the second level of screening, the two reviewers will examine the full articles that passed the first level of screening to determine if they meet the inclusion criteria. Both levels of screening will occur in duplicate.

Disagreements articles meeting the inclusion criteria will be resolved by consensus or in consultation with a third reviewer, if necessary. The rationale for excluding any of the studies will be documented, and a "Characteristic of Excluded Studies" will be made available for readers in the final review.

Data extraction

Researchers will independently collect all the relevant data from the articles that passed the first level of screening and record them on the review management software, Covidence.[30] A data extraction form will be used to collect information. In this form, the following information will be collected: (1) name of author(s), (2) year of publication, (3) country of origin, (4) study design (for example: cross-sectional study, or cohort study), (5) population (BD diagnosis and/or symptoms), (6) ADL assessment(s) used (7) type of ADL assessment used (self-reported, performance-based, or mixed), (8) ADLs examined and the respective level of functioning or independence reported and (9) any other study findings. During the critical appraisal, it will also be noted on the form if studies are at high risk of bias.

Critical appraisal

Following the screening, all included studies will be assessed for risk of bias in duplicate. For any observational studies, including case-control and cohort studies, they will be assessed based on the Newcastle-Ottawa Scale (NOS).[36] Cross-sectional studies will be assessed using

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a modified seven-question version of the NOS that was designed for cross-sectional studies.[37] For both versions, the raters will assign points for each of the quality parameters: selection, comparability, and outcome.[36] Studies that have less than 5 points in total are at high risk of biases for both the original and modified versions of the NOS.[37,38] Any studies that are at high risk of bias will be noted in the data extraction form. If any randomized control trials are included, the Joanne Briggs Institute (JBI) checklist for RCTs will be used to assess the risk of bias.[39]

Data synthesis

The review's findings will be reported based on the Preferred Reporting Items for Systematic reviews and Meta-Analyses, PRISMA.[28] A descriptive analysis will be used to report the participants' ADL functional independence across the different types of ADLs. A table will be created for the analysis called "Summary of Study Findings". The information in the table will include the following study properties: (1) name of author(s), (2) publication year, (3) country, (4) study design, (5) population (BD diagnosis and/or symptoms), (6) sample size, (7) assessment(s) used for ADL functioning, (8) outcome measure type(s) used (for example: self-reports, performance-based observations, or mixed), (9) reported ADL or functional outcomes per ADL/group of ADLs, and (10) other general findings of the articles.

A second table will then be created and titled "ADL Profile". This table be organized by separate ADL subheadings such as bathing, grooming, transportation, managing finances, and/or household management, depending on the included studies from the screening process. An ADL subheading will only be included if a minimum of two studies covered that specific ADL or the ADL domain. If appropriate, a meta-analysis will be used to group and quantitatively analyze individual studies similar in their assessments and the specific ADL outcomes.

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DISCUSSION

This protocol demonstrates a method for developing a systematic review examining current literature on how individuals manage their occupational functioning in terms of their ADLs with bipolar disorder. This protocol also presents a system for how it will screen for relevant articles, describe the trends found among the articles, and a means of assessing the quality of each included study.

This review will develop a resource that includes an ADL profile based on current primary literature. It will compile each of the different ADLs, such as bathing and dressing oneself, and report the findings for ADL independence or functioning in the basic and instrumental IADLs encountered from each of the included studies. Conducting this review will also identify gaps in occupation-based mental health literature and potentially evaluate current and the need for new ADL-based assessments for mental health disorders. Following the completion of the review, future avenues for research include developing best practice guidelines for targeting ADLs for patients with BD, as well as creating targeted interventions to improve independence and functioning in specific ADLs.

Ethics and dissemination

As this systematic review uses information from available primary literature, this review does not require ethics approval. This review's results will help identify the trends in ADLs that individuals with BD struggle with and can help health care clinicians develop targeted interventions for particular ADLs. We plan to disseminate the results in a peer reviewed journal and conferences that target occupational therapists and mental health workers.

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Competing interests: None declared.

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Appendix A

Table 1. Search strategies for each database**.**

Database	Search Strategy
Ovid MEDLINE	1. exp Activities of Daily Living/ 2. exp Self Care/ 3. exp Leisure Activities/ 4. ((“daily life” or “daily living” or instrument* or extended or leisure) adj3 activit*).tw,kf 5. (ADL* or BADL* or IADL* or EADL*).tw,kf 6. exp Functional Status/ 7. (function* adj3 (capacity or independen* or status or assess* or abilit*)).tw,kf 8. exp Bipolar Disorder/ 9. exp Cyclothymic Disorder/ 10. (bipolar adj3 (symptom* or diagnos* or disorder*)).tw,kf 11. manic adj2 depress*.tw,kf 12. 1 or 2 or 3 or 4 or 5 or 6 or 7 13. 8 or 9 or 10 or 11 14. 12 and 13
Embase	1. exp Activities of Daily Living/ 2. exp ADL disability/ 3. exp Daily life activity/ 4. exp Leisure/ 5. ((“daily life” or “daily living” or instrument* or leisure or extended) adj3 activit*).tw,kw 6. (ADL* or BADL* or IADL* or EADL*).tw,kw 7. exp Functional Status/ 8. exp Functional Assessment/ 9. (function* adj3 (capacity or independen* or status or assess* or abilit*)).tw,kw

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	10. exp Bipolar Disorder/ 11. exp Cyclothymic Disorder/ 12. (bipolar adj3 (symptom* or diagnos* or disorder*)).tw,kw 13. manic adj2 depress*.tw,kw 14. 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 15. 10 or 11 or 12 or 13 16. 14 and 15
APA PsycINFO	1. exp Activities of Daily Living/ 2. exp Self-Care/ 3. exp Leisure Time/ 4. ((“daily life” or “daily living” or instrument* or extended or leisure) adj3 activit*).ti,ab,id 5. (ADL* or BADL* or IADL* or EADL*).ti,ab,id 6. exp Functional Status/ 7. (function* adj3 (capacity or independen* or status or assess* or abilit*)).ti,ab,id 8. (bipolar adj3 (symptom* or diagnos* or disorder*)).ti,ab,id 9. exp Bipolar Disorder/ 10. exp Cyclothymic Disorder/ 11. manic adj2 depress*.ti,ab,id 12. 1 or 2 or 3 or 4 or 5 or 6 13. 7 or 8 or 9 14. 12 and 13
CINAHL	1. MH “Activities of Daily Living” 2. MH “Self Care” 3. MH “Leisure Activities” 4. TI ((“daily life” or “daily living” or instrument* or extended or leisure) N2 activit*) or AB ((“daily life” or “daily living” or instrument* or extended or leisure) N2 activit*)

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5. TI (self N2 care) OR AB (self N2 care)

6. TI (BADL* or IADL* or EADL*) OR AB (BADL* or IADL* or EADL*)

7. MH "Functional Status"

8. MH "Functional Assessment"

9. TI (function* N2 (capacity or independen* or status or assess* or abilit*)) OR AB (function* N2 (capacity or independen* or status or assess* or abilit*))

10. MH "Bipolar Disorder" or MH "Cyclothymic Disorder"

11. TI (bipolar N2 (symptom* or diagnos* or disorder*)) OR AB (bipolar N2 (symptom* or diagnos* or disorder*))

12. TI (manic N2 depress*) OR AB (manic N2 depress*)

13. 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9

14. 10 or 11 or 12

15. 13 or 14

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Reporting checklist for protocol of a systematic review and meta analysis.

Based on the PRISMA-P guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

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		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	6
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	N/A

Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	6
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	12
Amendments			
	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	12
Support			
Sources	#5a	Indicate sources of financial or other support for the review	12
Sponsor	#5b	Provide name for the review funder and / or sponsor	12
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	12
Introduction			

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Rationale	#6	Describe the rationale for the review in the context of what is already known	5-6
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	6
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	7
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	6
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	6-7
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	7
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	9

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Study records - data collection process	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	9
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	9
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	10-11
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	9-10
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	10
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	10-11
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	10-11

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Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	10-11
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	N/A
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	10-11

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Developing a profile of activities of daily living for bipolar disorder: A systematic review protocol and meta-analysis

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Developing a profile of activities of daily living for bipolar disorder: A systematic review protocol and meta-analysis

ABSTRACT

Introduction: Bipolar disorder is a severe mental health disorder that affects 2% of the adult population. Individuals with this disorder are at a higher risk for morbidity and functional difficulties. They may also experience significant challenges in their activities of daily living (ADLs). This systematic review will identify all available studies that examine subsets of ADLs that impact individuals with bipolar disorder.

Methods and analysis: Medline, Embase, CINAHL, and APA PsycINFO will be searched to identify observational studies that examined functioning, independence, or performance across a variety of ADLs. Title and abstract, full-text screening, and a risk of bias assessment will be conducted in duplicate. An overarching table that summarizes the level of functioning across different activities of daily living, or an 'ADL profile' will be developed, and if there is sufficient data, it will be separated based on the phases of bipolar disorder, such as manic/hypomanic, depressed, and euthymic.

Ethics and dissemination: As this systematic review uses information from previous literature, this review does not require ethics approval. This review will help identify the trends in daily activities that individuals struggle with and can help health care practitioners identify specific areas of need for support. We plan to disseminate the results in a peer reviewed journal and conferences targeting occupational therapists, and mental health clinicians.

Registration details: This systematic review has been registered on the PROSPERO International Prospective Register of systematic with registration number CRD42021255089.

Strengths and limitations of this study

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- This study employs a broad search strategy from multiple large databases, including searching reference lists of the included studies and employing controlled vocabulary terms for each database.
- Many studies currently only report bipolar disorder as a whole without subgroup analyses (for example, euthymic, depressive, manic), which may impact the interpretation of the results.
- Many articles related to measuring activities of daily living will be observational studies that could be prone to bias and confounding data, which may limit conclusions to the general bipolar disorder population.

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Developing a profile of activities of daily living for bipolar disorder: A systematic review protocol and meta-analysis

INTRODUCTION

Bipolar disorder (BD) is a severe and chronic mental health disorder characterized by recurrent mood episodes ranging from depressive to manic states and changes in energy levels.[1] There are two major forms of BD, including bipolar I disorder (BD-I), which involves mood disturbances as a manic episode with a hypomanic episode or depressive episode, whereas bipolar II disorder (BD-II) involves at least a depressive episode paired with a hypomanic episode.[1] When individuals are not experiencing mood disturbances, they would be in a euthymic state. BD typically presents in early adulthood and is estimated to have a lifetime prevalence of 0.6% for BD-I, and 0.4% for BD-II worldwide.[2] Lifetime prevalence of BD in the United States was estimated at 1.0% and 1.1% for BD I and II, respectively.[2] Individuals with BD also have an increased risk for comorbid mental health disorders and physical illnesses.[3] They may also be at high risk for premature mortality due to suicide and various medical illnesses such as circulatory or respiratory diseases than the general population.[4,5] From a meta-analysis by Hayes et al. (2015),[4] the standardized mortality ratio (SMR) was 2.05 for all-cause mortality, with suicide and unnatural death comprising the highest SMRs, at 14.44 and 7.42, respectively. BD also has high costs on society. In 2009, the annual cost of BD was estimated at $151 billion USD in the United States.[6] The majority of these costs are attributed to indirect costs, such as living in institutions and treatment related to functioning rather than direct treatment of symptoms.[6,7]

BD can greatly disrupt functioning in day to day activities across all life domains, including, but not limited to disruptions in basic functioning, work and social roles.[8–10] BD especially impacts a person's activities of daily living (ADLs), which includes any fundamental

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activities and routine sets of tasks that are relevant to independent functioning in everyday life, whether it involves self-care, homemaking, work, or leisure.[11] It is often distinguished into two types of ADLs: basic ADLs (BADLs), and instrumental or extended ADLs (IADLs or EADLs).[12,13] BADLs typically involve basic personal care activities such as mobilization, feeding, dressing, grooming, and toileting.[14] IADLs or EADLs typically encompasses higher level activities, such as managing transportation, handling finances, shopping, meal preparation, home maintenance, and managing medications.[13,14]

Original research studies looking at mental health and functioning have found that BD patients often have significantly decreased functioning and performance in ADLs. Numerous researching studies have found significantly decreased functioning in ADLs.[9,15–17] Even after remission from BD, studies report decreased ADL performance and participation in the BD population, where 30-60% of individuals never seeing a return to full functioning.[18] Decker et al. (2017) examined 43 patients in remission, and they found that functioning in daily life activities in these patients remained significantly impaired.[19] For 20-50% of those individuals, the dysfunctions manifested through increased physical effort, fatigue or disorganization.[19] Therefore, BD has a significant impact on daily life functioning, even when symptoms are minimal. Also, the recovery trends imply that absence of symptoms does not guarantee restored functioning; thus, individuals who are no longer receiving direct care for BD often continue to contribute to high societal costs.[6,7] Therefore, it is important to consider which areas of functioning in day-to-day activities are most impacted to further reduce societal costs.

In current literature, BD is often studied in terms of social relationships and employment. Studies have found that social functioning is impacted in various ways, such as decreased social interactions, and feelings of social discomfort.[20,21] They also may have impacted relationships

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with family, relatives, and marital relationships.[20,22] Likewise, BD appears to impact work employment, where there is a 40-60% rate of employment, and if employed, they may be more likely to take days off and underperform in their employment.[23,24] Nevertheless, people often need to participate in their day to day activities, such as household tasks, finances and personal care. If an individual's ability to manage their daily routine is impacted, then they may have difficulties maintaining their relationships and/or work roles.

Despite the general interest in ADL functioning across BD diagnoses, it is unknown which ADLs are most involved in BD, or if there are any trends in ADL functioning. A growing number of studies have examined specific ADLs individuals with BD have difficulties with.[19,25,26] There have also been primary studies examining how functioning across various ADL domains compare to work and social functioning.[9,16,27] However, there is no current systematic review that summarizes the findings of these studies in terms of the level of ADL functioning in subgroups of ADLs, such as household tasks, personal care, and managing finances. This systematic review will be conducted to help inform health care practitioners about the types and subgroups of ADLs that BD patients typically struggle with, and the extent of ADL independence or performance that BD patients typically have. Studying the level of independence for specific ADLs can help inform practical and therapeutic approaches in therapeutic settings by giving clinicians targeted areas for intervention planning. There are two questions that will be addressed in this upcoming review:

(1) What does the typical ADL profile look like for individuals with a bipolar disorder?

(2) Which ADLs do individuals with bipolar disorder typically have the most difficulties for functioning?

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METHODS

Study registration

This systematic review has been registered on the PROSPERO International Prospective Register of systematic with registration number CRD42021255089. The protocol has been prepared using the Preferred Reporting Items for Systematic reviews and Meta-Analyses for Protocols.[28]

Search strategy

All articles will be collected from the following databases: Ovid MEDLINE, Embase, APA PsycINFO, and CINAHL (Ebsco). A search strategy was developed with the aid of a health systems librarian. The search strategies were also reviewed using the PRESS 2015 Guideline Evidence-Based Checklist.[29] In general, the following search terms and variants of these terms will be used to acquire studies: “activities of daily living”, “functional independence”, and “bipolar disorder”. This search strategy incorporated Boolean searching, truncation, and adjacency. Controlled vocabulary terms were also included, such as Medical Subjects Headings (MeSH), CINAHL Subject Headings, and EMBASE’s Emtree. See Appendix A for the search strategy that was created for each database. Due to the risk of missing studies, searching reference lists of other included studies will identify relevant published studies that did not appear from the search strategy. The review management software, Covidence will collect all studies that pass the first and second level of screening.[30] Data collection is anticipated to start in March of 2022 and be completed by March 2023.

Inclusion criteria

Only original primary articles published in peer-reviewed journals will be included. Primary articles include any observational studies, including case-control, cross-sectional and

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cohort studies, but if relevant experimental studies are found, they will also be included. The article must be available in English. Non-English articles that appear from the search will be grouped under "Studies Awaiting Classification" in the PRISMA flow diagram to inform readers of the availability of other potentially relevant reports.[28] No restrictions will be placed on the article's year of publication.

With regards to the patient population, the study must include BD patients ages 18 or older. BD patients include individuals who were diagnosed one of the following based on the Diagnostic and Statistical Manual of Mental Disorders (DSM-V or DSM-IV): Bipolar I Disorder, Bipolar II Disorder, Cyclothymic Disorder, and unspecified bipolar and related disorder.[1] This study will also review studies with patients that have bipolar-specific states, such as euthymia or bipolar-related depressive symptoms, or bipolar-related cognitive deficits.

The inclusion criteria for the study also considers the functional performance or independence in activities of daily living (ADLs), including basic and instrumental/extended ADLs. Basic activities of daily living (BADLs) encompass any self-care activities, such as mobility, feeding, dressing, personal hygiene, or toileting. Instrumental ADLs may encompass transportation, managing finances, shopping, meal preparation, household management, or managing medications. To measure functional performance in ADLs, this review may include a variety of measures such as the UCSD Performance-Based Skills Assessment (UPSA),[31] Lawton IADL scale [32], Katz Index of Independence in Activities of Daily Living,[33] Performance Assessment of Self-Care Skills (PASS),[34] Functioning Assessment Short Test (FAST),[35] and any other relevant performance-based measures that have been previously validated in adult or older adult populations. Self-reported measures may also be used such as the

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Activities of Daily Living Questionnaire (ADL-Q).[19] Distinctions will be made between performance-based data and self-reported data.

Exclusion criteria

Studies will be excluded if they do not examine functional performance in terms of ADLs. They will also be excluded if ADL performance or independence is measured across multiple ADLs, meaning that it does not describe the participant's functioning in specific ADLs such as grooming or managing medications. Studies will also be excluded if they solely focus on work or employment or social functioning while disregarding BADLs or IADLs, as there are currently studies on social functioning, work, and relevance to BD.

Studies will be excluded if they report ADL performance or functioning globally across other mental health disorders, such as reporting ADL performance jointly with schizophrenia or major depressive disorder. Studies examining populations with BD and comorbidities will be excluded. Examples of comorbidities are neurological conditions, cancer diagnoses, substance abuse disorders, developmental delays, or other mental health disorders. Articles that study comorbidities and bipolar disorder without differentiating between the disorders are excluded, as studies that investigate comorbidities may also influence ADL performance. Likewise, studies that measure pharmacological or non-pharmacological treatments will be excluded as the purpose of this review is to examine ADLs rather than medical effects.

Screening and selection process

Study selection will occur in duplicate and conducted with two screening levels to determine if the articles meet the inclusion criteria. For the first level of screening, two reviewers will independently conduct the search strategy and review all the titles and abstracts from electronic database searches to screen for relevant studies. For the second level of screening, the

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two reviewers will examine the full articles that passed the first level of screening to determine if they meet the inclusion criteria. Both levels of screening will occur in duplicate. Disagreements articles meeting the inclusion criteria will be resolved by consensus or in consultation with a third reviewer, if necessary. The rationale for excluding any of the studies will be documented, and a "Characteristic of Excluded Studies" will be made available for readers in the final review.

Data extraction

Researchers will independently collect all the relevant data from the articles that passed the first level of screening and record them on the review management software, Covidence.[30] A data extraction form will be used to collect information. In this form, the following information will be collected: (1) name of author(s), (2) year of publication, (3) country of origin, (4) study design (for example: cross-sectional study, or cohort study), (5) population (BD diagnosis and/or symptoms), (6) average duration of the illness, (7) ADL assessment(s) used (8) type of ADL assessment used (self-reported, performance-based, or mixed), (9) ADLs examined and the respective level of functioning or independence reported and (10) any other study findings. During the critical appraisal, it will also be noted on the form if studies are at high risk of bias.

Critical appraisal

Following the screening, all included studies will be assessed for risk of bias in duplicate. For any observational studies, including case-control and cohort studies, they will be assessed based on the Newcastle-Ottawa Scale (NOS).[36] Cross-sectional studies will be assessed using a modified seven-question version of the NOS that was designed for cross-sectional studies.[37] For both versions, the raters will assign points for each of the quality parameters: selection, comparability, and outcome.[36] Studies that have less than 5 points in total are at high risk of

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biases for both the original and modified versions of the NOS.[37,38] Any studies that are at high risk of bias will be noted in the data extraction form. If any randomized control trials are included, the Joanne Briggs Institute (JBI) checklist for RCTs will be used to assess the risk of bias.[39]

Data synthesis

The review's findings will be reported based on the Preferred Reporting Items for Systematic reviews and Meta-Analyses, PRISMA.[28] A descriptive analysis will be used to report the participants' ADL functional independence across the different types of ADLs. A table will be created for the analysis called "Summary of Study Findings". The information in the table will include the following study properties: (1) name of author(s), (2) publication year, (3) country, (4) study design, (5) population (BD diagnoses studied, BD phases), (6) average onset and duration of the illness, (7) sample size, (8) assessment(s) used for ADL functioning, (9) outcome measure type(s) used (for example: self-reports, performance-based observations, or mixed), (10) reported ADL or functional outcomes per ADL/group of ADLs, and (11) other general findings of the articles.

A second table will then be created and titled "ADL Profile". This table be organized by separate ADL subheadings such as personal care, transportation, managing finances, and/or household management, depending on the included studies from the screening process. An ADL subheading will only be included if a minimum of two studies covered that specific ADL or the ADL domain. In addition, if there is sufficient data from the included studies, the level of independence for each ADL or ADL subgroup will be summarized across all phases of bipolar disorder, as well as the different phases of bipolar disorder: euthymic, depressed, and manic/hypomanic BD.

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A descriptive analysis will also be developed in a narrative format to compare findings from each study. The descriptive analysis would involve categorizing the different ADLs studied in primary literature. This descriptive analysis will be qualitatively grouped by similar ADLs, including self-care, productivity, and leisure. General findings will be described across the studies identified, and inferences will be drawn based on where ADL functioning is most impacted. In doing so, this information can help individuals and families better understand areas where the most support may be needed.

A meta-analysis will also be included if there is sufficient statistical information from the included studies. If common assessments were used across multiple studies, or if there were common statistical measures used across studies such as relative risks or odds ratios, a meta-analysis will be conducted with this information. A random effects approach will be employed, where effect sizes and variance will be calculated. A weighted mean of these effect sizes will then be generated. As there will be clinical and methodological differences between the studies, heterogeneity between studies will be tabulated with the I^2 statistic. The I^2 statistic will be calculated as per the Cochrane Handbook for Systematic Reviews.[40] Statistical analyses will be conducted using R 4.02 and the "metafor" (version 2.4–0) package. [41-43]

Patient and public involvement

There was no patient or public involvement in this study.

DISCUSSION

This protocol demonstrates a method for developing a systematic review examining current literature on how individuals manage their occupational functioning in terms of their ADLs with bipolar disorder. This protocol also presents a system for how it will screen for relevant articles, describe the trends found among the articles, and a means of assessing the

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quality of each included study. It also considers the different phases of BD such as euthymia, manic/hypomanic episodes and depressive episodes, which may also play a crucial role in impacting ADL functioning.

In creating this review, a resource will be developed to integrate current primary literature. The resource will compile various ADLs, such as household tasks, personal care, and work, and report the functioning across different ADLs and ADL subgroups encountered from each of the included studies. Conducting this review will also identify gaps in occupation-based mental health literature and potentially evaluate current and the need for new ADL-based assessments for mental health disorders. Following the completion of the review, future avenues for research include developing best practice guidelines for targeting ADLs for patients with BD, as well as creating targeted interventions to improve independence and functioning in specific ADLs.

Ethics and dissemination

As this systematic review uses information from available primary literature, this review does not require ethics approval. This review's results will help identify the trends in ADLs that individuals with BD struggle with and can help health care clinicians develop targeted interventions for particular ADLs. We plan to disseminate the results in a peer reviewed journal and conferences that target occupational therapists and mental health workers.

Contribution statement:

BNK, AA, and FS planned the study protocol. AA drafted the manuscript for this review protocol. Author BNK and FS edited the initial protocol manuscript drafts. BNK, AA, FS, JS, and HC all contributed to revising, reviewing, and approving of the manuscript. BNK prepared the final manuscript for submission and submitted the manuscript for review.

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Competing interests:

None declared.

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Appendix A

Table 1. Search strategies for each database**.**

Database	Search Strategy
Ovid MEDLINE	1. exp Activities of Daily Living/ 2. exp Self Care/ 3. exp Leisure Activities/ 4. ((“daily life” or “daily living” or instrument* or extended or leisure) adj3 activit*).tw,kf 5. (ADL* or BADL* or IADL* or EADL*).tw,kf 6. exp Functional Status/ 7. (function* adj3 (capacity or independen* or status or assess* or abilit*)).tw,kf 8. exp Bipolar Disorder/ 9. exp Cyclothymic Disorder/ 10. (bipolar adj3 (symptom* or diagnos* or disorder*)).tw,kf 11. manic adj2 depress*.tw,kf 12. 1 or 2 or 3 or 4 or 5 or 6 or 7 13. 8 or 9 or 10 or 11 14. 12 and 13
Embase	1. exp Activities of Daily Living/ 2. exp ADL disability/ 3. exp Daily life activity/ 4. exp Leisure/ 5. ((“daily life” or “daily living” or instrument* or leisure or extended) adj3 activit*).tw,kw 6. (ADL* or BADL* or IADL* or EADL*).tw,kw 7. exp Functional Status/ 8. exp Functional Assessment/ 9. (function* adj3 (capacity or independen* or status or assess* or abilit*)).tw,kw

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	10. exp Bipolar Disorder/ 11. exp Cyclothymic Disorder/ 12. (bipolar adj3 (symptom* or diagnos* or disorder*)).tw,kw 13. manic adj2 depress*.tw,kw 14. 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 15. 10 or 11 or 12 or 13 16. 14 and 15
APA PsycINFO	1. exp Activities of Daily Living/ 2. exp Self-Care/ 3. exp Leisure Time/ 4. ((“daily life” or “daily living” or instrument* or extended or leisure) adj3 activit*).ti,ab,id 5. (ADL* or BADL* or IADL* or EADL*).ti,ab,id 6. exp Functional Status/ 7. (function* adj3 (capacity or independen* or status or assess* or abilit*)).ti,ab,id 8. (bipolar adj3 (symptom* or diagnos* or disorder*)).ti,ab,id 9. exp Bipolar Disorder/ 10. exp Cyclothymic Disorder/ 11. manic adj2 depress*.ti,ab,id 12. 1 or 2 or 3 or 4 or 5 or 6 13. 7 or 8 or 9 14. 12 and 13
CINAHL	1. MH “Activities of Daily Living” 2. MH “Self Care” 3. MH “Leisure Activities” 4. TI ((“daily life” or “daily living” or instrument* or extended or leisure) N2 activit*) or AB ((“daily life” or “daily living” or instrument* or extended or leisure) N2 activit*)

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5. TI (self N2 care) OR AB (self N2 care)

6. TI (BADL* or IADL* or EADL*) OR AB (BADL* or IADL* or EADL*)

7. MH "Functional Status"

8. MH "Functional Assessment"

9. TI (function* N2 (capacity or independen* or status or assess* or abilit*)) OR AB (function* N2 (capacity or independen* or status or assess* or abilit*))

10. MH "Bipolar Disorder" or MH "Cyclothymic Disorder"

11. TI (bipolar N2 (symptom* or diagnos* or disorder*)) OR AB (bipolar N2 (symptom* or diagnos* or disorder*))

12. TI (manic N2 depress*) OR AB (manic N2 depress*)

13. 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9

14. 10 or 11 or 12

15. 13 or 14

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Reporting checklist for protocol of a systematic review and meta analysis.

Based on the PRISMA-P guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

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Moher D, Shamseer L, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart LA. Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2015 statement. Syst Rev. 2015;4(1):1.

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		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	6
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	N/A

Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	6
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	12
Amendments			
	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	12
Support			
Sources	#5a	Indicate sources of financial or other support for the review	12
Sponsor	#5b	Provide name for the review funder and / or sponsor	12
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	12
Introduction			

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Rationale	#6	Describe the rationale for the review in the context of what is already known	5-6
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	6
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	7
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	6
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	6-7
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	7
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	9

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Study records - data collection process	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	9
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	9
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	10-11
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	9-10
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	10
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	10-11
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	10-11

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Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	10-11
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	N/A
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	10-11

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